David Le Chan

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Education

M.S. in Electrical and Computer Engineering (ECE) Expected May 2027
Carnegie Mellon University (CMU) | Pittsburgh, PA

- GPA: /4.00

B.S. in Electrical and Computer Engineering (ECE) Expected May 2026
Carnegie Mellon University (CMU) | Pittsburgh, PA

- GPA: 3.8/4.0; *University Honors, 6x College of Engineering Dean's List Recipient*

Work Experience

Hardware Development Engineering Intern May 2026 - August 2026
Annapurna Labs (Amazon AWS), Machine Learning Acceleration | Austin, TX

- Developed, validated, and deployed PCIe connectivity solutions
- Designed and implemented qualification tests for rack-level switch sleds
- Improved automation capabilities with bash and lua scripts
- Identified and resolved design issues impacting over 100,000 production units

FPGA & Electrical Engineering Intern May 2025 - August 2025
KLA Corporation, Broadband Plasma Semiconductor Inspection | Milpitas, CA

- Upgraded FPGA firmware for high-speed lossless image compression on KLA's flagship wafer inspection tool line
- Implemented subsystems via Verilog and Vivado IP Integrator, verifying functionality through Questa simulation
- Deployed designs on PCIe-based Alveo accelerator cards, performing place-and-route optimization and hardware-level validation to quantify performance and ensure reliability

Undergraduate Research Assistant May 2024 - Present
IO Harness Project, CMU ECE | Pittsburgh, PA

- Designing a standardized harness to reduce infrastructure redevelopment work for IC tapeouts; first prototype fabricated on TSMC 180 nm under active testing
- Architecting system features and writing SystemVerilog RTL for I2C, UART, and SPI communication blocks

Power Electronics & Programming Intern June 2022 - January 2024
Tau Motors | Redwood City, CA

- Prototyped power circuits for wound-field electric motors, including PCB layout, assembly, and bench testing
- Developed custom Python-based inventory management software and systems to accelerate hardware iterations

Leadership and Projects

Hybrid FFT/NTT Accelerator SoC January 2026 - Present

- End-to-end design of a novel hybrid accelerator that supports both FFT and NTT operations through hardawre reuse, culminating in a 28nm tapeout
- Integrated our hybrid accelerator with a RISC-V core to create a test SoC in SpinalHDL
- Designed a custom cocotb-based test orchestrator for multi-level verification at all design stages

Head Teaching Assistant (TA), Introduction to ECE (18-100) January 2025 - January 2026

- Led a 40 TA team to foster an emotionally-safe environment where 180 first-year students can explore ECE, form lasting friendships, and develop strong engineering habits
- Continuously redesigned lab curriculum to give students hands-on experience with real-world design workflows
- Established automated Python scripts and feedback systems to streamline course operations, reducing

administrative overhead and empowering TAs to focus on mentorship and teaching quality